FIRST-COLLISION HURWITZ DYNAMICS ON MINIMAL CYCLE FACTORIZATIONS: SHARP TAILS, FIXED CENSUS, AND EXACT FIBRES

ANONYMOUS

Abstract. Let $\mathcal{F}_n$ be the minimal transposition factorizations of the long cycle $(1\,2\,\cdots\,n)$. At each epoch, find the first adjacent pair of transpositions having the same lower endpoint and apply the right Hurwitz move there; hold if no such pair exists. Although each local move is classical, this state-dependent scheduler creates a terminating finite map. We prove that successive update positions increase strictly, giving the sharp tail bound $n-2$. Via the classical lower-endpoint bijection with parking functions and Pollak's circular model, the fixed-state count is $(n-1)^{n-2}$. Independently, we classify every predecessor of every labelled target by one inverse Hurwitz test. The maximum one-step fibre is $n-1$, attained uniquely by the canonical adjacent-transposition factorization. Exact exhaustion through $n = 8$ checks 280,392 states and reveals a stronger binomial law for complete scheduler histories; because an all-n bijection is not yet proved, that law is stated only as a conjecture. Hurwitz actions, factorization–parking correspondences, and ordinary Prüfer enumerations receive zero contribution credit. The owner gate is `OWNER_RED_AMBER/HOLD_EXTERNAL`.

1. Convention, map, and subtraction

Fix $n \geq 2$ throughout. Permutations compose in the standard order: in $\sigma\rho$, the permutation ρ acts first. Put $c_n = (1\,2\,\cdots\,n)$ and

$$\mathcal{F}_n = \{(\tau_1, \ldots, \tau_{n-1}) : \tau_i \text{ is a transposition and } \tau_1 \cdots \tau_{n-1} = c_n\}. \tag{1}$$

For a transposition (a, b) written with $a < b$, set $\ell((a, b)) = a$. The right Hurwitz move at i is

$$H_i(\ldots, \tau_i, \tau_{i+1}, \ldots) = (\ldots, \tau_{i+1}, \tau_{i+1}\tau_i\tau_{i+1}, \ldots). \tag{2}$$

It preserves the product.

Define $T_n : \mathcal{F}_n \to \mathcal{F}_n$ by choosing the least $i \in \{1, \ldots, n-2\}$ with

$$\ell(\tau_i) = \ell(\tau_{i+1}) \tag{3}$$

and applying H_i; if no such i exists, fix the state. The product, Hurwitz orientation, numeric endpoint order, and least-index convention are part of the definition. Reversing only the long-cycle orientation does not preserve the dynamics.

Minimal long-cycle factorizations, their Cayley-tree count, and Hurwitz actions are classical [2, 3]. The lower-endpoint map

$$(\tau_1, \ldots, \tau_{n-1}) \longmapsto (\ell(\tau_1), \ldots, \ell(\tau_{n-1})) \tag{4}$$

is the classical bijection from $\mathcal{F}_n$ to parking functions of length $n-1$ [4, 5]. All of this receives zero contribution credit. The residual claims concern the adaptive first-collision scheduler and its target-resolved inverse graph.

Campion Loth and Rattan give a deterministic bijection built from ordered strings of conditional Hurwitz moves in their study of star and monotone factorizations [1]; in particular, equality of lower endpoints appears in one case of their local construction. That conditional Hurwitz-string machinery also receives zero contribution credit here. Its move convention, monotone/string order-change scheduler, reversible-bijection objective, and theorem output differ from the present repeated map, which always selects the first adjacent lower-endpoint collision and studies its tails and target-resolved fibres. This distinction is an exact object separation, not a novelty claim.

Internally, direct Hurwitz actions have repeatedly been owner-killed, and P181 uses first-descent prefix reversal on ordinary permutations. The present map is not a braid-group orbit census, a 0-Hecke action, or a prefix reversal. Indeed its conditional local operators fail the braid relation already on $((1, 4), (1, 3), (1, 2))$. A bounded owner search found the classical ingredients above but no literal scheduler theorem package. This non-hit is not evidence of novelty, priority, completeness, or freedom to operate.

Key words and phrases. Hurwitz action, transposition factorization, parking function, finite dynamical system, transient, fibre.

2. Strictly advancing collisions

For $f \in \mathcal{F}_n$, let $i_1, i_2, \ldots$ be its successive update positions.

Theorem 2.1 (scheduler normal form and sharp tail)**.** *Every nonempty update history is strictly increasing:*

$$1 \le i_1 < i_2 < \cdots \le n-2. \tag{5}$$

Consequently every recurrent state is fixed and every tail is at most $n-2$. For $n \ge 3$ the bound is sharp; one witness is

$$((1,n),(1,2),(2,3),\ldots,(n-2,n-1)), \tag{6}$$

whose history is $1,2,\ldots,n-2$ and whose terminal state is $((1,2),(2,3),\ldots,(n-1,n))$. For $n=2$, the sole factorization $((1,2))$ is fixed, so the same sharp bound equals zero.

Proof. At an active position write $\tau_i = (a,b)$ and $\tau_{i+1} = (a,c)$, where $a < b, c$. The two transpositions are distinct: otherwise they cancel and the n-cycle would be expressed using fewer than $n-1$ transpositions. The update replaces them by

$$(a,c),(b,c). \tag{7}$$

The new lower endpoints at positions $i, i+1$ are $a, \min(b,c)$, and the second is strictly larger than a. All earlier lower endpoints are unchanged, including the endpoint at position i. Since i was the first old equality, no equality is created before i and the equality at i disappears. The next update position is therefore strictly larger. This proves (5), termination, and the upper bound.

For $n \ge 3$, the first move in (6) changes the initial two factors to $(1,2),(2,n)$. The next collision is at position two. Inductively the factor (j,n) meets $(j,j+1)$, and the Hurwitz move replaces them by $(j,j+1),(j+1,n)$. The stated history and terminal chain follow. □

The proof supplies a pointwise certificate: the history is a subset of $[n-2]$ written in increasing order, and the depth is its cardinality.

3. Exact fixed-state census

Put $N = n-1$. A parking function is a word $a \in [N]^N$ whose increasingly sorted entries $b_1 \le \cdots \le b_N$ satisfy $b_i \le i$.

Theorem 3.1 (fixed factorizations)**.** *The fixed states of T_n are exactly the factorizations whose lower-endpoint word has no equal adjacent letters, and*

$$|\operatorname{Fix}(T_n)| = (n-1)^{n-2}. \tag{8}$$

Proof. The characterization is the stopping rule. Under (4), it remains to count parking functions $a_1 \cdots a_N$ with $a_i \ne a_{i+1}$ for $i < N$.

Use Pollak's circular parking model on $N+1$ spots [6]. There are $(N+1)N^{N-1}$ circular preference words with adjacent entries unequal: choose the first preference freely and each later one in N ways. Simultaneous translation modulo $N+1$ preserves the inequalities, every translation orbit has size $N+1$, and each orbit has exactly one representative that becomes an ordinary parking function after placing the empty spot at the end. Division by $N+1$ gives N^{N-1}, which is (8). □

4. Every-target inverse Hurwitz atlas

Fix a target $y = (\sigma_1, \ldots, \sigma_{n-1})$. Let

$$j(y) = \min\{j : \ell(\sigma_j) = \ell(\sigma_{j+1})\}, \tag{9}$$

with sentinel value $j(y) = n-1$ when the set is empty. For $i < j(y)$, write $\sigma_i = (a,b)$ with $a < b$. Call i reverse-admissible when

$$\sigma_{i+1} = (b,c) \text{ for some } c > a, \tag{10}$$

where the transposition is unordered, so the condition means that it contains b and its other endpoint exceeds a.

Theorem 4.1 (complete labelled fibre)**.** *Every target has exact indegree*

$$\operatorname{indeg}(y) = \mathbf{1}_{\{j(y)=n-1\}} + \#\{i < j(y) : i \textit{ is reverse-admissible}\}. \tag{11}$$

For each counted i, the unique nonself predecessor is $H_i^{-1}y$. Moreover,

$$\max_{y\in\mathcal{F}_n} \operatorname{indeg}(y) = n-1, \tag{12}$$

and the unique maximizer is the canonical chain

$$((1,2),(2,3),\dots,(n-1,n)). \tag{13}$$

For $n = 2$, this chain is the sole target and its self-fibre has indegree $1 = n-1$.

Proof. Suppose a nonself source updates at i to y. Positions before i are unchanged, while the source equality at i is removed, so necessarily $i < j(y)$. Inverting (2) gives the pair

$$(\sigma_i\sigma_{i+1}\sigma_i, \sigma_i). \tag{14}$$

For its two lower endpoints to agree, σ_{i+1} must contain the upper endpoint b of $\sigma_i = (a,b)$, and its other endpoint c must exceed a. The coincident-transposition case would cancel and is impossible. Thus (10) is necessary. Conversely it turns (14) into $(a,c),(a,b)$; positions before i retain no collision, so the scheduler selects exactly i. This proves the atlas and uniqueness of each inverse.

A fixed target contributes itself and at most one inverse at each of the $n-2$ positions; a nonfixed target contributes fewer. Hence $\operatorname{indeg}(y) \le n-1$. Every adjacent pair of (13) is reverse-admissible, so it attains equality. Equality forces a fixed target with every position admissible. If $\sigma_i = (a_i,b_i)$, admissibility makes both endpoints of σ_{i+1} exceed a_i, whence $a_{i+1} > a_i$. A strictly increasing parking function of length $n-1$ is necessarily $(1,2,\dots,n-1)$, and the classical lower-endpoint bijection then forces the unique factorization (13). ☐

Iterating (11) gives an exact finite reverse dynamic program for every higher fibre; no uniform closed form is asserted.

5. A verified history law left open

The exact controls expose a stronger pattern. We isolate it to prevent finite evidence from silently becoming a theorem.

Conjecture 5.1 (history-set law)**.** *For every $I \subseteq [n-2]$,*

$$\#\{f \in \mathcal{F}_n : \operatorname{Hist}(f) = I\} = (n-1)^{n-2-|I|}. \tag{15}$$

Equivalently, the number of states of exact depth t would be

$$\binom{n-2}{t}(n-1)^{n-2-t}. \tag{16}$$

The verifier proves (15) only for $2 \le n \le 8$ by exhaustion. A separate Cayley-tree stream also checks $n = 9$. The histogram matches the occurrence-set law of a distinguished letter in a Prüfer word, but tested standard factorization–tree bijections do not identify the scheduler history with those positions. No all-n bijection is presently known here. Accordingly (15), the resulting unique deepest-state claim, and any derived basin formula receive no theorem status.

6. Controls and limitations

The paper-local standard-library verifier generates the full Hurwitz orbit from (13), checks its Cayley cardinality and product, the lower-endpoint parking bijection, every orbit, every history mask through $n = 8$, and every target fibre against (11). It also checks the orientation-sensitive convention: only the frozen product/order is covered. These computations are regression pressure, not proof or novelty evidence.

The paper does not claim a new Hurwitz action, conditional Hurwitz-string bijection, parking-function bijection, or tree code. It does not cover inverse-cycle conventions, other reflection groups, nonminimal factorizations, random schedulers, or the conjectural all-n history law. External circulation remains unauthorized under `HOLD_EXTERNAL`.

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